Glints, nightside light, satellite belts, timing steps and occulter shapes: photometric technosignature searches in Kepler, TESS and Cepheid timing

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Abstract

Artificial objects around other stars could show in their photometry as glints of reflected light at an orbital period, as light from a planet's dark side, as a belt of satellites crossing the star with its planet, as a timing signal written into a regular clock, or as an occulter whose shape no planet has. We search the Kepler and TESS archives for each. Folded at the periods of 816 Kepler and 794 TESS planet candidates whose hosts show three or more brief flashes in 252239 and 317274 star-days of short-cadence photometry, the flashes gather at one phase no more often than chance predicts, and the flashes in 2149 TESS 20 s light curves of stars within 30 pc recur at no period beyond chance. The nightsides of 63 ultra-short-period rocky planets emit no light above 3σ, with a median limit of 102 ppm of the star's flux. Belts of satellites 3 to 90 planetary radii from 461 Kepler planets add no shoulders to their transits once each transit is timed and measured from its own light curve, with a median 3σ limit of 117 ppm six radii out. Of 1810 Kepler planet candidates with 20 or more transit times, none shows a phase step held from one transit on that a sinusoidal timing variation or chance does not explain, nor do the times of maximum of 2939 LMC Cepheids over 14.6 years. Those of 1116 Kepler and 255 TESS rejected candidates whose light curves fold to a stable profile are eclipsing binaries, stellar variability, heartbeat binaries and smooth dips of constant depth.

Keywords: Technosignatures, Techniques: photometric, Planets and satellites: detection, Stars: variables: Cepheids, Occultations

1. Introduction

The photometric archives of Kepler [1] and TESS [2] hold years of precise light curves of hundreds of thousands of stars, and several kinds of artificial object would leave marks in them. Flat surfaces in orbit, such as mirrors or solar panels, reflect sunlight specularly and appear as brief glints repeating at the orbital period [3]. Artificial lighting on a planet's nightside would show as light before and after transit that the planet's own heat does not explain [4]. An occulter of non-planetary shape would give a transit profile no limb-darkened disc produces [5, 6]. And a civilization could signal by shifting the phase of a regular clock: the transits of a body it controls, or the pulsations of a Cepheid it perturbs [8].

Where these ideas have been tested, it has been on individual objects. This paper applies each to the whole of an archive, with nulls drawn from the same light curves, so that the number of signals expected by chance is known.

1.1. Contributions

1. Searches for glints at every Kepler and TESS planet-candidate period and at any period in the TESS 20 s light curves of the nearest stars.
2. A uniform limit on the nightside light of 63 ultra-short-period rocky planets.
3. A search of the transits of 461 Kepler planets for the shoulders of satellite belts.
4. Searches for held phase steps in the transit times of 1810 Kepler candidates and in the pulsation timing of 2939 LMC Cepheids.
5. Shape tests of 1116 Kepler and 255 TESS rejected candidates.

2. Searches

Glints at planet periods. For the 765 hosts of Kepler planet candidates with short-cadence data and the 6355 hosts of TESS objects of interest (TOIs) that are neither false positives nor false alarms [9], each light curve, divided by a running median of 30 (Kepler) or 62 (TESS) minutes, gives flashes: brightenings five robust deviations high lasting one to three cadences, with quiet flux on either side. For every candidate on a star with three or more flashes, the flashes are folded at its period, and the busiest window of 2.5 cadences in phase is compared with the same statistic for 400 sets of flash times drawn from the star's own good cadences. The per-cadence scatter of the detrended light curves has a median of 780 ppm in the Kepler short-cadence sample and 1600 ppm in the TESS two-minute sample, so a flash must reach about 0.4 and 0.8 per cent of the star's flux. The quiet flux required on either side, four cadences within two robust deviations, passes a flash in white noise with probability $0.954^8 = 0.69$, and the package's end-to-end test of the detector on synthetic Kepler cadences recovers 0.62 of isolated flashes at ten deviations, so the searches see about two thirds of the flashes above their thresholds. A flat mirror of area A and reflectivity ρ that reflects the star's disc toward the observer adds a glint of $\rho A/\pi R_\star^2$ of the star's flux, whatever its distance from the star, so those thresholds correspond to perfect reflectors of 6×10^{15} and $8\times10^{15}\,\mathrm{m}^2$, 80 000 to 90 000 km on a side, around a star of solar radius. A mirror whose orientation follows its orbit sweeps its reflection across the observer in $R_\star P/2\pi a$, 1.5 to 6 hours at 0.05 to 1 au from a solar radius, longer than the running median and the one-to-three-cadence window, so what the search covers is a reflector that spins, whose beam crosses the observer in minutes and recurs at the orbital phase where the geometry repeats.

Glints at any period. The TESS 20 s light curves of Gaia DR3 stars within 30 pc give flashes in the same way, over a 10 minute running median. For a star with three or more flashes, every period implied by a pair of flashes, and its fractions down to a fifth, is tested for flashes within 40 s of the predicted times, against the same count for flash times drawn from the star's good cadences. The 20 s cadences scatter by a median of 3000 ppm, so a flash must reach 1.5 per cent of the flux, a reflector of $3\times10^{15}\,\mathrm{m}^2$ around an M dwarf of 0.35 solar radii.

Nightside light. A rocky planet's nightside faces us just before and after transit and is hidden in secondary eclipse, and below about 1500 K it emits next to nothing in the TESS band. Every TOI with a period below 1 d and a radius below $2\,R_\oplus$ is folded over its two-minute sectors and fitted outside transit and eclipse with a sinusoid and its first harmonic; the nightside flux is the fit at phase zero less the mean flux in eclipse.

Satellite belts. A belt of satellites k planetary radii from a planet, such as one at its synchronous radius [7], crosses the star with the planet and dims it for kR_p/v before the planet's first contact

and as long after its last, where v is the planet's speed across the star. The long-cadence light curves of the 1243 hosts of confirmed Kepler planets brighter than Kepler magnitude 15, detrended by a running median that skips each transit and its shoulders and with the system's other transits masked, are folded on Holczer et al.'s measured transit times where they exist [10] and on the catalogue ephemeris otherwise. For k from 3, beyond the Roche limit inside which natural rings lie, to 90 or half the planet's Hill radius, the flux from 18 minutes beyond each contact to kR_p/v beyond it is compared with the flux just outside, for both shoulders together and each alone, against the same statistic at 40 random phases away from transit and eclipse. Every test above $z = 4$ is measured again with each transit's time and the transit's duration fitted from the light curve, against a null whose phases are detrended with the same gap as the transit, and a shoulder must exceed $z = 5$ in both.

Phase steps in transit times. Holczer et al. give the time of every transit of 2599 Kepler candidates [10]. A perturbing planet moves transit times sinusoidally, tides and precession move them smoothly, and the spacecraft's quarterly rolls shift them at quarter boundaries; a step held from one transit on needs an impulsive change of the orbit, which natural dynamics makes only in a close encounter and which a civilization that moves a body, or that times an artificial occulter, would make at will. The held step is the simplest symbol such a clock could carry, and the one that a search without a model of the message can test. For each candidate with 20 or more good times, the observed minus calculated times are fitted with a linear ephemeris, with a step held from one transit on (six or more transits each side) and with the best sinusoid. A step counts when it beats the sinusoid by 10 in χ^2, has a false-alarm probability below 0.01 against 300 permutations of the residuals, and lies more than two days from a quarter boundary.

Phase steps in Cepheid timing. Learned et al. proposed that a civilization could advance a Cepheid's pulsation and write information into its times of maximum [8]. Poleski's times of maximum of LMC Cepheids from MACHO (1992–1999) and OGLE (1997–2008) [11] give each of 2939 stars a median of 82 maxima over 14.6 years; each star's timing is fitted with a quadratic ephemeris with offsets between sources, then with a step held from one cycle on or with a companion's light-travel orbit of 1 to 30 years, and random walks of the star's own excess scatter give the null.

Occulter shapes. The Kepler DR25 candidates flagged only as not transit-like [12], 1116 of them, and the 255 TESS false positives and false alarms at $T \leq 11.5$ whose dispositions name no binary, blend or offset, are folded at their periods with stellar variability removed, fitted with a symmetric trapezoid, and measured for their departure from it, their ingress-egress asymmetry, their number of dips and the agreement between halves of the data and between odd and even events. A periodic, stable profile that no trapezoid describes and that has no secondary eclipse is listed.

3. Results

Glints. The Kepler light curves hold 3634 flashes in 252239 star-days; of the 816 candidates on stars with three or more, 9 gather their flashes at one phase at a false-alarm probability below 0.01, against at most 8.2 by chance, and 2 reach the 1/401 floor of the draws, against 2.0. The TESS light curves hold 8710 flashes in 317274 star-days, and none of the 794 TOIs on stars with three or more falls below a false-alarm probability of 0.01, against at most 7.9 by chance. The 2149 TESS 20 s light curves, 1.13 million star-hours, hold 19695 flashes, 0.42 a star-day, mostly cosmic rays and instrumental events; in the 1232 stars with three or more, the best periodic family of each reaches $p \leq 0.01$ in 6 stars against 12.3 expected for uniform p, and $p \leq 0.05$ in 38 against 61.6.

Nightside light. For the 63 planets the nightside flux divided by its error has a median of −0.33 and a spread of 0.87, and none exceeds +3; the highest, TOI 4567.01 at 2.9, has an equally high level in eclipse, an offset of the eclipse level rather than light from the planet. The median 3σ limit is 102 ppm of the star's flux. That is far more than the starlight the planets receive: a planet of radius R_p at distance a intercepts $(R_p/2a)^2$ of its star's light, 0.5 to 33 ppm for these 63 (median 6.8 ppm), so the limits lie 3 to 100 times (median 21) above all the starlight a nightside could re-emit, and constrain only light powered by something other than the star: at the median limit, 4×10^{21} W, with a range of 2×10^{20} to 10^{23} W, which is 10^4 to 10^6 times humanity's output. A nightside lit by its star's own energy, at or below a few parts per million, needs a precision these light curves do not reach.

Satellite belts. The 1243 stars give 1522 tests of 461 planets. Eight exceed $z = 5$ with both shoulders above $z = 2$, where no null phase reaches $z = 5$, and every test above $z = 4$ falls below $z = 5$ when measured again: with each transit's own time and fitted duration, against the null detrended as the transit is, the largest of the 22 planets on the 16 stars concerned reaches $z = 3.3$ (KOI-1353.02). The excesses come from transit-timing variations that a linear ephemeris smears across the contacts (KOI-134.01, whose transits wander by about ten hours, falls from $z = 28.7$ against the matched null to 1.0 on its own times, and KOI-1474.01 from 16.8 to −0.2), from catalogue durations shorter than the transit (KOI-620.01 lasts 7.4 h against the catalogue's 5.8 h), and from the running median's gap at the transit, which the matched null reproduces (KOI-1375.01 falls from 6.8 to 3.0). A shoulder six planetary radii out is excluded at 3σ down to 117 ppm of the star's flux for the median planet and 22 ppm for the best.

Phase steps. Of the 1810 Kepler candidates, 8 steps beat the sinusoid by 10 in χ^2, one at a quarter boundary, where steps are set aside. The strongest of the other seven follow the long-period timing variations that Holczer et al. list (KOI-12.01, KOI-824.01 and KOI-2310.01), and the rest are low-signal timings at false-alarm probabilities of 0.0012 to 0.0020, of which two to four are expected among the trials. Among the Cepheids, four steps pass the tests: three fall at the start of OGLE-II's first season (HJD 2450467), where the data change source, and one on a date that 15 other stars share, so steps at a season's start or on a date other stars share are excluded and none remains.

Occulter shapes. Of the Kepler rejects, 129 fold to periodic, stable profiles at S/N 15 or more without a secondary eclipse: 81 are Kepler eclipsing binaries, among them the one strongly asymmetric profile (KOI-3716.01), and the 48 others are rotational and pulsational variability, heartbeat binaries, or smooth V- and U-shaped dips of constant depth (KOI-3198, KOI-8060, KOI-8240 and KOI-7698). Of the TESS rejects, 63 are periodic and stable over three or more events; the strongest departures are stellar variability (TOI 1447.01), an eclipse on a blue supergiant (TOI 796.01), an eclipse whose depth changes between sectors (TOI 7247.01) and a grazing eclipsing binary (TOI 1090.01).

4. Related work

Transit anomalies. Arnold showed how the transits of artificial occulters differ from planets' [5], the $\hat{G}$ search set out the signatures of transiting megastructures [6], and Breakthrough Listen searched 218 Kepler systems transit by transit for anomalies [13].

Glints, nightsides and clocks. Specular reflections from artificial surfaces were proposed as a technosignature by Jaiswal [3], artificial illumination detectable by its light curve by Loeb and Turner [4], and phase coding of Cepheids by Learned et al. [8], whose idea Hippke et al. tested on one

Kepler RR Lyrae star [14]. Socas-Navarro showed that a belt of satellites at a planet's synchronous radius would add shoulders to its transits [7]. We find no earlier archive-wide search for glints at planet periods, for nightside light across the ultra-short-period planets, for the shoulders of satellite belts, or for held phase steps in transit or Cepheid timing.

5. Conclusion

No glint repeats at a planet-candidate period in Kepler or TESS, or at any period in the fast light curves of the nearest stars, beyond what chance gives; no ultra-short-period rocky planet shows light from its nightside above a hundredth of a per cent of its star, twenty times what its star could power; no Kepler planet's transits carry the shoulders of a satellite belt once each transit is timed and measured from the light curve; no transit or Cepheid timing holds a phase step that natural timing variations, instrument changes or chance do not explain; and the rejected candidates with stable, non-planetary profiles are eclipsing binaries, variable stars and smooth dips.

The glint searches apply to flashes lasting one to three cadences at five robust deviations, 0.4 to 1.5 per cent of the star's flux, which perfect reflectors of 3 to $8 \times 10^{15}\,\mathrm{m}^2$ would give, and to reflectors whose beam crosses the observer in minutes; the nightside limit, 102 ppm, is 21 times the starlight the median planet intercepts and bounds only light of a non-stellar source above about $4 \times 10^{21}\,\mathrm{W}$; the belt search applies to belts 3 to 90 planetary radii out whose shoulders outlast 18 minutes beyond contact, and the timing searches to single held steps. A glint shorter than a cadence or longer than three, a reflector smaller than a planet, a signal at a period other than a candidate's in the longer-cadence data, a belt inside the Roche limit, or timing coded in many small shifts is not constrained.

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References

[1] W. J. Borucki, D. Koch, G. Basri, et al., Kepler planet-detection mission: introduction and first results, *Science* **327**, 977 (2010).

[2] G. R. Ricker, J. N. Winn, R. Vanderspek, et al., Transiting Exoplanet Survey Satellite, *J. Astron. Telesc. Instrum. Syst.* **1**, 014003 (2015).

[3] B. Jaiswal, Specular reflections from artificial surfaces as technosignature, *Astrobiology* **23**, 291 (2023).

[4] A. Loeb and E. L. Turner, Detection technique for artificially illuminated objects in the outer solar system and beyond, *Astrobiology* **12**, 290 (2012).

[5] L. F. A. Arnold, Transit light-curve signatures of artificial objects, *Astrophys. J.* **627**, 534 (2005).

[6] J. T. Wright, K. M. S. Cartier, M. Zhao, D. Jontof-Hutter and E. B. Ford, The $\hat{G}$ search for extraterrestrial civilizations with large energy supplies. IV. The signatures and information content of transiting megastructures, *Astrophys. J.* **816**, 17 (2016).

[7] H. Socas-Navarro, Possible photometric signatures of moderately advanced civilizations: the Clarke exobelt, *Astrophys. J.* **855**, 110 (2018).

[8] J. G. Learned, R.-P. Kudritzki, S. Pakvasa and A. Zee, The Cepheid galactic internet, *Contemp. Phys.* **53**, 113 (2012).

[9] N. M. Guerrero, S. Seager, C. X. Huang, et al., The TESS objects of interest catalog from the TESS prime mission, *Astrophys. J. Suppl.* **254**, 39 (2021).

[10] T. Holczer, T. Mazeh, G. Nachmani, et al., Transit timing observations from Kepler. IX. Catalog of the full long-cadence data set, *Astrophys. J. Suppl.* **225**, 9 (2016).

[11] R. Poleski, Period changes of LMC Cepheids in the OGLE and MACHO data, *Acta Astron.* **58**, 313 (2008).

[12] S. E. Thompson, J. L. Coughlin, K. Hoffman, et al., Planetary candidates observed by Kepler. VIII. A fully automated catalog with measured completeness and reliability based on data release 25, *Astrophys. J. Suppl.* **235**, 38 (2018).

[13] A. Zuckerman, J. Davenport, S. Croft, A. Siemion and I. de Pater, The Breakthrough Listen search for intelligent life: detection and characterization of anomalous transits in Kepler lightcurves, arXiv:2312.07903 (2023).

[14] M. Hippke, J. G. Learned, A. Zee, et al., Pulsation period variations in the RRc Lyrae star KIC 5520878, *Astrophys. J.* **798**, 42 (2015).